

Chapter 16 - ServiceNow (SNOW) Ticketing Tool

Introduction to ServiceNow (SNOW)

ServiceNow (SNOW) is an IT Service Management (ITSM) tool used by companies to:

-  Create tickets
-  Track issues
-  Assign work to teams
-  Monitor SLA
-  Manage IT operations

It works like a **digital complaint system** for IT problems.

Real-Life Example

Imagine a hospital  :

- Patient = User
- Problem = IT Issue
- Reception = Ticket Creation
- Doctor = IT Engineer
- Treatment = Troubleshooting
- Discharge = Ticket Closure

ServiceNow manages this entire process professionally.

L1 Level Process Change

Earlier:

- IT Support team created and assigned tickets.

Now:

-  IT Coordinator creates the ticket.
- IT Coordinator assigns the ticket to the correct team.

This improves:

-  Proper categorization
-  Better tracking
-  Accountability

How to Create a Ticket in ServiceNow (Step-by-Step)

Go to:

New → Incident

Now fill the following fields carefully 

1 Company

Which company does the user belong to?

 Used for:

- Identifying client
 - Applying correct SLA
 - Reporting & billing
-

2 Caller

Person who raised the ticket.

Why important?

- For communication
 - For accountability
 - For updates
-

3 Email ID

User's official email.

Used for:

- Ticket updates
 - Notifications
 - Closure confirmation
-

4 Phone Number

For urgent issues or quick clarification.

5 Location

User's physical or office location.

Why?

- Field engineer dispatch
 - Time zone management
 - Network site identification
-

Category (Very Important ⚠)

This defines issue type.

Common Categories:

-  Database
-  Hardware
-  Network
-  Security
-  Software
-  Storage
-  Inquiry
-  Request

Category helps in:

- Assigning correct team
- Reporting analytics
- SLA calculation

Subcategory

More specific classification.

Example:

Category: Hardware

Subcategory: Laptop / Mouse / Printer

This increases accuracy.

8 Configuration Item (CI) – Very Important

CI means:

Which device/server/router has issue?

Example:

- User Laptop Name
- Router Name
- Firewall ID

This is linked to **CMDB (Configuration Management Database)**.

 Helps track:

- Device history
 - Past incidents
 - Warranty details
-

9 Channel

How user contacted IT:

-  Phone
-  Email
-  Walk-in
-  Chat
-  Self-service portal
-  Virtual Agent
-  External Event

Used for:

- Reporting
 - Performance tracking
-

10 State

Ticket status.

Common States:

-  New
 -  Active
 -  Awaiting Customer
 -  Awaiting Evidence
 -  Awaiting Vendor
 -  Awaiting Problem
 -  Resolved
 -  Closed
-

1 1 Assignment Group

Which team should handle it?

Example IT Infra has:

- Cyber Security Team
- Network Team
- IT Support
- DevOps Team

Assignment Group decides:

Where ticket goes.

1 2 Assigned To

Specific engineer handling the ticket.

1 3 Impact, Urgency & Priority (Very Important)

Priority depends on:

Impact + Urgency = Priority

Scenario Example

User 1:

Laptop not starting → Work completely stopped.

User 2:

Mouse not working → Work possible via keyboard.

 User 1 gets higher priority.

Priority Levels

- P1 – Critical
 - P2 – High
 - P3 – Medium
 - P4 – Low
-

SLA (Service Level Agreement)

Defines resolution time.

General Example:

Priority SLA

P1 1 Hour

P2 1 Hour

P3 24 Hours

Company-wise standards differ.

Cyber Security SLA

In Cyber Security:

- ⚡ MTTR = 5 minutes
 - Must assign quickly
-

MTTR (2 Types)

1 Minimum Time to Response

Time to acknowledge and assign ticket.

2 Minimum Time to Resolve

Time to completely fix issue.

Both are critical metrics.

Description Field

Detailed explanation of issue.

Should include:

- What happened
- When happened
- Error message
- Steps tried

This helps faster resolution.

Work Notes

Used by engineer.

Contains:

- Troubleshooting steps
- Commands used

- Findings
- Root cause

⚠ Work notes are internal.

Additional Comments

Visible to customer.

Used for:

- Updates
 - Clarifications
 - Closure confirmation
-

After Submission

Once submitted:

- 🎫 Ticket number generated
 - 📧 Email sent to user
 - 🔄 Updates sent automatically
-

VIP Ticket Rule

If 4 normal tickets + 1 VIP ticket:

👉 Always prioritize VIP ticket.

VIP impact is higher.



As Network Engineer – How You Pick Ticket

- 1 Check Assignment Group
 - 2 Open ticket
 - 3 Review Description
 - 4 Check Priority
 - 5 Accept / Assign to yourself
 - 6 Change state to Active
 - 7 Start troubleshooting
-



Router Access Process

⚠ You cannot connect router directly.

Company provides:

-  Physical Laptop
 -  Virtual Machine (VM)
 -  PuTTY Application
-



Steps to Connect Router

- 1 Open VM
 - 2 Open PuTTY
 - 3 Enter Router IP
 - 4 Connect via SSH
-



NSD Database

NSD = Network System Database

Contains:

- Router Name
- Public IP

- Device Details
- Site Info

Before connecting:

👉 Get device IP from NSD.



How Ticket Appears in List View

After creation, you see:

- 📄 Ticket Number
 - 🔥 Priority
 - 👤 Assigned To
 - 🏢 Company
 - 📝 Short Description
 - 📋 Configuration Item
 - 📅 Updated Date
 - 📅 Opened Date
 - 👥 Assignment Group
 - 👤 Caller
 - 📶 Channel
 - 🔄 State
 - ⌚ SLA Due
 - ⏳ Activity Due
 - 📅 Follow-up Date
-

Ticket Lifecycle

New → Active → Awaiting → Resolved → Closed

Important:

- Resolved ticket can reopen (within SLA period).
 - Closed ticket cannot reopen.
 - Usually closes after 2–3 days automatically.
-

Why We Don't Close Immediately?

Because:

User may say issue not fixed.

So:

Resolved → Wait → Auto close.

How Ticket Assigning Works in Company

Flow:

User → IT Coordinator → Assignment Group → Engineer → Resolution → QA → Closure

This ensures:

- Transparency
 - Documentation
 - SLA compliance
-

Search Bar

You can search:

- Incident number
 - User name
 - CI name
-

Filters

Used to filter:

- My Tickets
 - P1 Tickets
 - Overdue SLA
-

Dashboards

Used by managers to monitor:

- MTTR
 - SLA Breach
 - Team performance
-

CMDB Integration

Each device linked to:

- Owner
 - Location
 - History
 - Change records
-

Escalation Process

If SLA about to breach:

- Escalate to manager
 - Change priority if needed
 - Notify stakeholders
-

Final Summary

ServiceNow is:

-  Ticket Management System
-  SLA Tracking Tool
-  IT Governance Platform
-  Work Documentation System

As Network Engineer you must:

- Understand priority
 - Respect SLA
 - Update work notes properly
 - Communicate clearly
 - Follow ticket lifecycle
-



DETAILED SUMMARY



What is ServiceNow?

ServiceNow (SNOW) is an **IT Service Management (ITSM)** tool used to:

- 📄 Create incidents
- 🛠️ Track issues
- 👤 Assign tasks
- ⌚ Monitor SLA
- 📊 Measure performance
- 📁 Maintain documentation

It acts as a **digital complaint and workflow system for IT operations.**



L1 Process Change



Earlier:

IT Support team created & assigned tickets.



Now:

- IT Coordinator creates ticket
- IT Coordinator assigns correct team



Benefits:

- Better categorization
 - Clear accountability
 - Improved SLA tracking
 - Better reporting
-



Ticket Creation – Important Fields (Very Important for Interview)



1 Company

Identifies which client/user belongs to which organization.

Used for:

- SLA application
- Billing
- Reporting



2 Caller

Person who raised the ticket.

Important for:

- Communication
- Accountability
- Updates



3 Email ID

Used for:

- Notifications
- Closure confirmation
- Updates



4 Phone Number

Used for urgent clarification.



5 Location

Helps in:

- Field engineer dispatch
 - Site identification
 - Time zone management
-

Category (Very Important)

Defines issue type.

Examples:

-  Network
-  Database
-  Hardware
-  Security
-  Software
-  Storage
-  Inquiry
-  Request

Used for:

- Assigning correct team
 - SLA calculation
 - Reporting
-

Subcategory

More specific issue type.

Example:

Category: Hardware

Subcategory: Printer

8 Configuration Item (CI)

CI = Affected device or asset.

Example:

- Router name
- Firewall ID
- Laptop hostname

Connected to:

 CMDB (Configuration Management Database)

Helps track:

- Device history
 - Past incidents
 - Warranty
-

9 Channel

How ticket was raised:

- Phone
- Email
- Walk-in
- Chat
- Portal
- Virtual Agent

Used for reporting performance.

10 State

Ticket status lifecycle:

-  New

-  Active
-  Awaiting Customer
-  Awaiting Vendor
-  Awaiting Evidence
-  Resolved
-  Closed

1 1 Assignment Group

Which team handles ticket?

Examples:

- Network Team
- Cyber Security Team
- IT Support
- DevOps

1 2 Assigned To

Specific engineer handling issue.

1 3 Impact + Urgency = Priority (Very Important)

Formula:

Impact + Urgency = Priority

Priority Levels

-  P1 – Critical
-  P2 – High

-  P3 – Medium
 -  P4 – Low
-

Example:

Laptop not working → Work stopped → High Impact → P1

Mouse not working → Work possible → Lower Impact → P3

SLA (Service Level Agreement)

Defines resolution time.

Example:

Priority SLA

P1	1 Hour
P2	1 Hour
P3	24 Hours

Cyber Security SLA

-  MTTR = 5 Minutes
 - Immediate assignment required
-

MTTR – Two Types

1 Minimum Time to Response
Time to acknowledge ticket.

2 Minimum Time to Resolve
Time to completely fix issue.

Description Field

Must include:

- What happened
- When happened
- Error message
- Steps already tried

Clear description = Faster resolution.

Work Notes

Internal notes.

Includes:

- Commands used
- Troubleshooting steps
- Root cause

⚠ Not visible to customer.

Additional Comments

Visible to user.

Used for:

- Updates
 - Clarifications
 - Closure confirmation
-

After Submission

System:

- Generates ticket number
- Sends email notification

- Tracks SLA automatically
-

VIP Ticket Rule

If:

4 Normal Tickets + 1 VIP Ticket

 VIP ticket gets priority.

Network Engineer Ticket Handling Process

- 1 Check Assignment Group
 - 2 Open ticket
 - 3 Review description
 - 4 Check priority
 - 5 Accept ticket
 - 6 Change state to Active
 - 7 Start troubleshooting
-

Router Access Process

You cannot directly connect to router.

Company provides:

- Physical laptop
 - Virtual Machine (VM)
 - PuTTY software
-

Router Connection Steps

- 1 Open VM
- 2 Open PuTTY
- 3 Enter router IP
- 4 Connect via SSH

NSD (Network System Database)

Contains:

- Router name
- Public IP
- Device details
- Site information

Used before accessing router.

Ticket Lifecycle

New → Active → Awaiting → Resolved → Closed

Important:

- Resolved ticket can reopen.
 - Closed ticket cannot reopen.
 - Usually auto closes after 2–3 days.
-

Dashboards

Managers monitor:

- MTTR
 - SLA breaches
 - Team performance
 - Open vs closed tickets
-

CMDB Integration

CMDB stores:

- Device owner
- Location

- History
 - Change records
-

Escalation Process

If SLA about to breach:

- Escalate to manager
- Notify stakeholders
- Increase priority if needed



Q & A

Q 1 What is ServiceNow?

Answer:

ServiceNow is an ITSM tool used for ticket management, SLA tracking, and IT operations workflow automation.

Q 2 What is an Incident?

An unplanned interruption in IT service.

Q 3 What is SLA?

Service Level Agreement defining resolution time for tickets.

Q 4 What is MTTR?

Minimum Time to Response or Resolve.

Q 5 Difference between Work Notes and Additional Comments?

Work Notes → Internal

Additional Comments → Customer visible

Q 6 What is Configuration Item (CI)?

The affected device or asset linked in CMDB.

Q 7 What is CMDB?

Configuration Management Database storing asset details and relationships.

Q 8 How is Priority calculated?

Impact + Urgency = Priority

Q 9 What is VIP Ticket?

Ticket raised by high-level user. It gets priority.

Q 10 Explain Ticket Lifecycle.

New → Active → Awaiting → Resolved → Closed

Q 1 1 Can a closed ticket reopen?

No. Only resolved tickets can reopen.

Q 1 2 What is Assignment Group?

Team responsible for resolving issue.

Q 1 3 What happens if SLA breaches?

Escalation to manager and reporting impact.

Q 1 4 Why category is important?

It ensures correct team assignment and accurate reporting.

Q 1 5 How do you handle a P1 ticket?

Immediately acknowledge, assign, start troubleshooting, update regularly, resolve within SLA.

Q 1 6 What is NSD?

Network System Database containing router IP and site details.

Q 1 7 How do you connect to router in corporate environment?

Using VM → PuTTY → SSH connection.

Q 1 8 Why don't we close ticket immediately?

Because user may report issue not fixed. We wait before closure.

Q 1 9 What dashboards show?

SLA compliance, MTTR, team performance.
